Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-9 Emerged Berm 2% - Test 1.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.242 Hz*
Freq. upper bound:	*4.844 Hz*

Freq. Domain Analysis

Overlap (%) of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*32*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*40*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *41.5%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1221*	*0.05069*
Peak wave period T_P (s):	*1.882*	*2*
Energy wave period $T_{-1,0}$ (s):	*1.773*	*1.9*
Mean wave period $T_{0,1}$ (s):	*1.591*	*1.552*
Mean wave period $T_{0,2}$ (s):	*1.477*	*1.307*
Spectral width (Broadness):	*0.7428*	*0.8727*
Spectral width (Narrowness):	*0.4*	*0.6399*

Time Domain Analysis

Overall amplitude reflection coefficient: *41.3%*

	Incident	**Reflected**
Number of waves:	991	955
Mean wave height H_m (m):	*0.0783*	*0.03381*
Mean wave period T_m (s):	*1.677*	*1.741*
Sig. wave height H_s (m):	*0.1222*	*0.05239*
$T_{H_{1/3}}$ (s):	*1.89*	*2.035*
H_{max} (m):	*0.2151*	*0.1038*
T_{Hmax} (s):	*2.12*	*2.31*
$H_{1/10}$ (m):	*0.1537*	*0.06432*
$H_{1/50}$ (m):	*0.1853*	*0.07622*
$H_{1/100}$ (m)	*0.1953*	*0.07985*
$H_{1/250}$ (m):	*0.2066*	*0.08683*
$H_{0.1\%}$ (m):	*Not enough data*	*Not enough data*
$H_{1\%}$ (m):	*0.1829*	*0.07399*
$H_{2\%}$ (m):	*0.1711*	*0.06984*
$H_{10\%}$ (m)	*0.1326*	*0.05552*
Groupiness factor GF:	*1.108*	*1.023*
Skewness b1:	*0.7349*	*0.09585*
Kurtosis b2:	*3.498*	*2.953*

Department of Civil Engineering

Aalborg University

Project title	
Drawn by	
Date	*11. mrt 2026*
Remarks	
Page	*1 of 7*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-9 Emerged Berm 2% - Test 1.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.242 Hz*
Freq. upper bound:	*4.844 Hz*

Freq. Domain Analysis

Overlapping (%) of timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*34*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*40*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *42.2%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1325*	*0.05586*
Peak wave period T_P (s):	*2.065*	*2.065*
Energy wave period $T_{-1,0}$ (s):	*1.823*	*1.965*
Mean wave period $T_{0,1}$ (s):	*1.611*	*1.546*
Mean wave period $T_{0,2}$ (s):	*1.478*	*1.268*
Spectral width (Broadness):	*0.7757*	*0.8834*
Spectral width (Narrowness):	*0.4329*	*0.6974*

Time Domain Analysis

Overall amplitude reflection coefficient: *42.1%*

	Incident	**Reflected**
Number of waves:	1020	1015
Mean wave height H_m (m):	*0.08674*	*0.03774*
Mean wave period T_m (s):	*1.754*	*1.762*
Sig. wave height H_s (m):	*0.1345*	*0.05783*
$T_{H_{1/3}}$ (s):	*1.979*	*2.074*
H_{max} (m):	*0.2221*	*0.1005*
T_{Hmax} (s):	*1.632*	*1.821*
$H_{1/10}$ (m):	*0.1655*	*0.06998*
$H_{1/50}$ (m):	*0.1901*	*0.08243*
$H_{1/100}$ (m)	*0.1971*	*0.08785*
$H_{1/250}$ (m):	*0.2034*	*0.09613*
$H_{0.1\%}$ (m):	*0.2217*	*0.1005*
$H_{1\%}$ (m):	*0.1914*	*0.07956*
$H_{2\%}$ (m):	*0.1801*	*0.07506*
$H_{10\%}$ (m)	*0.1459*	*0.06191*
Groupiness factor GF:	*1.097*	*0.9643*
Skewness b1:	*0.815*	*0.07326*
Kurtosis b2:	*3.505*	*2.764*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-9 Emerged Berm 2% - Test 1.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.227 Hz*
Freq. upper bound:	*4.531 Hz*

	Gauge 1	Gauge 2	Gauge 3
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Freq. Domain Analysis

Overlap (FFT) timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*35*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*42*
Hilbert filter length:	*2048*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *43.8%*

	Incident	Reflected
Sig. wave height H_{m0} (m):	*0.1442*	*0.06315*
Peak wave period T_P (s):	*2.207*	*2.207*
Energy wave period $T_{-1,0}$ (s):	*1.895*	*2.07*
Mean wave period $T_{0,1}$ (s):	*1.644*	*1.56*
Mean wave period $T_{0,2}$ (s):	*1.49*	*1.247*
Spectral width (Broadness):	*0.8061*	*0.889*
Spectral width (Narrowness):	*0.4668*	*0.7527*

Time Domain Analysis

Overall amplitude reflection coefficient: *43.8%*

	Incident	Reflected
Number of waves:	1029	1019
Mean wave height H_m (m):	*0.09448*	*0.04302*
Mean wave period T_m (s):	*1.795*	*1.813*
Sig. wave height H_s (m):	*0.1464*	*0.06468*
$T_{H_{1/3}}$ (s):	*2.059*	*2.138*
H_{max} (m):	*0.2399*	*0.107*
T_{Hmax} (s):	*1.801*	*2.176*
$H_{1/10}$ (m):	*0.1792*	*0.07684*
$H_{1/50}$ (m):	*0.1987*	*0.09003*
$H_{1/100}$ (m)	*0.2056*	*0.09465*
$H_{1/250}$ (m):	*0.2164*	*0.0997*
$H_{0.1\%}$ (m):	*0.2391*	*0.1069*
$H_{1\%}$ (m):	*0.1951*	*0.08984*
$H_{2\%}$ (m):	*0.1891*	*0.08273*
$H_{10\%}$ (m)	*0.1604*	*0.06853*
Groupiness factor GF:	*1.065*	*0.9314*
Skewness b1:	*0.8358*	*0.02543*
Kurtosis b2:	*3.426*	*2.649*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-9 Emerged Berm 2% - Test 1.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.219 Hz*
Freq. upper bound:	*4.375 Hz*

Freq. Domain Analysis

Overlap (FFT) timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*37*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*44*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *44.4%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1542*	*0.06847*
Peak wave period T_P (s):	*2.286*	*2.286*
Energy wave period $T_{-1,0}$ (s):	*1.927*	*2.124*
Mean wave period $T_{0,1}$ (s):	*1.642*	*1.537*
Mean wave period $T_{0,2}$ (s):	*1.474*	*1.202*
Spectral width (Broadness):	*0.8181*	*0.8908*
Spectral width (Narrowness):	*0.4923*	*0.797*

Time Domain Analysis

Overall amplitude reflection coefficient: *44.4%*

	Incident	**Reflected**
Number of waves:	1034	1032
Mean wave height H_m (m):	*0.1035*	*0.04762*
Mean wave period T_m (s):	*1.84*	*1.842*
Sig. wave height H_s (m):	*0.1585*	*0.06989*
$T_{H_{1/3}}$ (s):	*2.101*	*2.182*
H_{max} (m):	*0.2455*	*0.1052*
T_{Hmax} (s):	*2.378*	*2.588*
$H_{1/10}$ (m):	*0.1895*	*0.08343*
$H_{1/50}$ (m):	*0.2124*	*0.09341*
$H_{1/100}$ (m)	*0.2184*	*0.09627*
$H_{1/250}$ (m):	*0.225*	*0.1002*
$H_{0.1\%}$ (m):	*0.2447*	*0.1051*
$H_{1\%}$ (m):	*0.2111*	*0.09261*
$H_{2\%}$ (m):	*0.2034*	*0.08796*
$H_{10\%}$ (m)	*0.1709*	*0.07417*
Groupiness factor GF:	*1.075*	*0.9*
Skewness b1:	*0.9259*	*-0.001745*
Kurtosis b2:	*3.517*	*2.615*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-9 Emerged Berm 2% - Test 1.experiment.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.219 Hz*
Freq. upper bound:	*4.375 Hz*

Freq. Domain Analysis

Overlapping timeseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*38*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*44*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *44%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1598*	*0.07028*
Peak wave period T_P (s):	*2.286*	*2.37*
Energy wave period $T_{-1,0}$ (s):	*1.959*	*2.152*
Mean wave period $T_{0,1}$ (s):	*1.645*	*1.481*
Mean wave period $T_{0,2}$ (s):	*1.46*	*1.137*
Spectral width (Broadness):	*0.8354*	*0.8881*
Spectral width (Narrowness):	*0.5183*	*0.834*

Time Domain Analysis

Overall amplitude reflection coefficient: *44.2%*

	Incident	**Reflected**
Number of waves:	1037	1072
Mean wave height H_m (m):	*0.1094*	*0.04937*
Mean wave period T_m (s):	*1.896*	*1.834*
Sig. wave height H_s (m):	*0.1652*	*0.07236*
$T_{H_{1/3}}$ (s):	*2.181*	*2.238*
H_{max} (m):	*0.2457*	*0.106*
T_{Hmax} (s):	*1.667*	*1.998*
$H_{1/10}$ (m):	*0.1954*	*0.08556*
$H_{1/50}$ (m):	*0.2224*	*0.09741*
$H_{1/100}$ (m)	*0.2331*	*0.1006*
$H_{1/250}$ (m):	*0.2439*	*0.1039*
$H_{0.1\%}$ (m):	*0.2456*	*0.106*
$H_{1\%}$ (m):	*0.2222*	*0.09679*
$H_{2\%}$ (m):	*0.2069*	*0.09187*
$H_{10\%}$ (m)	*0.1757*	*0.07677*
Groupiness factor GF:	*1.06*	*0.886*
Skewness b1:	*0.9555*	*-0.03389*
Kurtosis b2:	*3.493*	*2.599*

Reflection Analysis

Input Parameters

Generel

Data file:	*26-3-9 Emerged Berm 2% - Test 1.experiment overtop (4).txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.328 Hz*
Freq. upper bound:	*6.563 Hz*

Freq. Domain Analysis

Overlap of subseries:	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*25*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*29*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *34.6%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.08232*	*0.02851*
Peak wave period T_P (s):	*1.524*	*1.524*
Energy wave period $T_{-1,0}$ (s):	*1.442*	*1.454*
Mean wave period $T_{0,1}$ (s):	*1.367*	*1.291*
Mean wave period $T_{0,2}$ (s):	*1.318*	*1.175*
Spectral width (Broadness):	*0.5889*	*0.7907*
Spectral width (Narrowness):	*0.2743*	*0.4566*

Time Domain Analysis

Overall amplitude reflection coefficient: *34.6%*

	Incident	**Reflected**
Number of waves:	966	1017
Mean wave height H_m (m):	*0.05189*	*0.01808*
Mean wave period T_m (s):	*1.358*	*1.289*
Sig. wave height H_s (m):	*0.08013*	*0.0283*
$T_{H_{1/3}}$ (s):	*1.492*	*1.54*
H_{max} (m):	*0.135*	*0.04665*
T_{Hmax} (s):	*1.466*	*1.739*
$H_{1/10}$ (m):	*0.09779*	*0.035*
$H_{1/50}$ (m):	*0.1156*	*0.04162*
$H_{1/100}$ (m)	*0.1217*	*0.04402*
$H_{1/250}$ (m):	*0.1277*	*0.04575*
$H_{0.1\%}$ (m):	*Not enough data*	*0.04664*
$H_{1\%}$ (m):	*0.1128*	*0.04134*
$H_{2\%}$ (m):	*0.1063*	*0.03792*
$H_{10\%}$ (m)	*0.08425*	*0.03011*
Groupiness factor GF:	*0.9747*	*0.9664*
Skewness b1:	*0.3306*	*0.238*
Kurtosis b2:	*2.943*	*2.935*

Reflection Analysis

Input Parameters

Generel

Data file:	*260-3-9 Emerged Berm 2% - Test 1.experimentversuch.txt*
Sample frequency:	*128 Hz (Downsampled to 64 Hz)*
Water depth at Gauge 1:	*0.365 m (seabed slope 1:200)*
Length scale (Prototype/Model):	*1*
Skipped lines	*Header: 5 Start: 1000 End: 0*

Bandpass filtering

Freq. lower bound:	*0.281 Hz*
Freq. upper bound:	*5.625 Hz*

Freq. Domain Analysis

Overlap (FFT):	*20%*
Cosine taper width:	*20%*
Data No. in FFT block:	*4096*
Number of FFT blocks:	*28*

Time Domain Analysis

Output timeseries at x-coordinate:	*0.3333 m*
Min. points between downcrossings:	*34*
Hilbert filter length:	*2048*

	Gauge 1	*Gauge 2*	*Gauge 3*
Channel number:	*2*	*3*	*4*
Calibration function:	*1.0*X*	*1.0*X*	*1.0*X*
X-coordinate:	*0.0 m*	*0.3 m*	*0.7 m*

Frequency Domain Analysis

Overall amplitude reflection coefficient: *38.4%*

	Incident	**Reflected**
Sig. wave height H_{m0} (m):	*0.1068*	*0.04102*
Peak wave period T_P (s):	*1.778*	*1.778*
Energy wave period $T_{-1,0}$ (s):	*1.633*	*1.699*
Mean wave period $T_{0,1}$ (s):	*1.512*	*1.446*
Mean wave period $T_{0,2}$ (s):	*1.431*	*1.262*
Spectral width (Broadness):	*0.6857*	*0.845*
Spectral width (Narrowness):	*0.3393*	*0.5595*

Time Domain Analysis

Overall amplitude reflection coefficient: *38.3%*

	Incident	**Reflected**
Number of waves:	970	989
Mean wave height H_m (m):	*0.06793*	*0.02658*
Mean wave period T_m (s):	*1.533*	*1.502*
Sig. wave height H_s (m):	*0.1052*	*0.04253*
$T_{H_{1/3}}$ (s):	*1.708*	*1.785*
H_{max} (m):	*0.1758*	*0.0865*
T_{Hmax} (s):	*2*	*2.002*
$H_{1/10}$ (m):	*0.1305*	*0.05306*
$H_{1/50}$ (m):	*0.1578*	*0.06574*
$H_{1/100}$ (m)	*0.165*	*0.07171*
$H_{1/250}$ (m):	*0.171*	*0.07713*
$H_{0.1\%}$ (m):	*Not enough data*	*Not enough data*
$H_{1\%}$ (m):	*0.1555*	*0.06515*
$H_{2\%}$ (m):	*0.1467*	*0.05732*
$H_{10\%}$ (m)	*0.114*	*0.04568*
Groupiness factor GF:	*1.047*	*1.065*
Skewness b1:	*0.5339*	*0.1894*
Kurtosis b2:	*3.187*	*3.223*

Department of Civil Engineering

Aalborg University

Project title	
Drawn by	
Date	*11. mrt 2026*
Remarks	
Page	*7 of 7*